Universal-Atlas Pandrosion for Smale-17 Stress Systems

fixed chart geometry, finite-slope cubic correction, and the exact complexity gap remaining after benchmark 304

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Abstract

This paper records and formalizes the 304 Pandrosion engine: a fixed universal atlas of complex projective cells followed by strict finite-slope cubic Halley correction. The point of 304 is different from the previous 301 engine. In 301 the chart generator is a single Mobius–Riemann–Thales flow whose parameters react to root history. In 304 the geometric atlas is fixed in advance for each dimension: simplex, hypercube, golden-projective, reciprocal shell, and frozen Mobius/Riemann cells. The input polynomial system does not choose a scalar best chart; it only activates cells through predicate tests for finite residual, local descent, equal-value separation, and logarithmic scale stability. This makes 304 a cleaner candidate for a probabilistic basin-access theorem.

We prove the unconditional algebraic pieces: the exact telescopic finite-slope identity, polynomial arithmetic cost per trial when the number of cells/probes/epochs is polynomially bounded, local cubic convergence under a compatible finite-slope basin condition, and the standard implication that inverse-polynomial basin access gives average-polynomial time and high-probability success. We do not claim the missing global basin-access theorem is proved. Instead we isolate it precisely as the remaining Smale-17 obstruction for 304.

The large 304 diagnostic benchmark completed 528/528 runs on Kostlan stress cases, recovered 21940/21952 requested roots, certified 21666/21940 roots by alpha-lite tests, produced 17283/21940 strong roots, and achieved median residual $1.02 \cdot 10^{-13}$ after a median 0.110 seconds per root. Compared with 301 on the same suite, 304 finds the same number of roots but certifies substantially more of them, at about 2.75× median runtime cost.

Proof ledger. The finite-slope identity, local cubic theorem, per-trial arithmetic bound, expected-time theorem, and repetition amplification theorem are proved below. The global statement that a fixed universal atlas hits compatible approximate-zero basins with inverse-polynomial probability for Kostlan input is stated as Problem 5.4; it remains the missing theorem, not a hidden consequence of the benchmark.

1 Smale-17 target and the 304 change

Let $F = (F_1, \ldots, F_n) : \mathbb{C}^n \to \mathbb{C}^n$ be a square dense polynomial system of degree at most d. Write

$$N = \binom{n+d}{d}$$

for the number of affine monomials per equation. In the Kostlan model $\mathcal{K}(n, d)$ the coefficients are complex Gaussian with the usual binomial variance normalization. Smale's seventeenth problem asks for an algorithm that returns an approximate zero of a random input in average time polynomial in the dense input size, comparable here to nN.

The 304 engine is intended as a non-homotopy route. It does not track paths in coefficient space. A trial performs:

(i) choose a cell from a fixed universal atlas in $\mathbb{C}^n$;

(ii) test only predicates using F: finite residual, bounded logarithmic scale, local residual descent under fixed probes, and non-negligible equal-value gap;

(iii) choose a cyclic admissible representative, not a scalar minimizer;

(iv) apply finite local start balancing;

(v) build the exact finite-slope matrix $Q_F(a,b)$ from an endpoint pair (a,b);

(vi) solve the strict cubic correction

$$Q_F(a,b)\delta_1 = -F(a), \qquad Q_F(a,b)\delta_2 = -\frac{1}{2}H_F(a;b;\delta_1,\delta_1),$$

where the second defect is estimated by symmetric finite probes;

(vii) line-search only along $\delta = \delta_1 + \delta_2$ and validate the residual.

The raw finite-slope step δ_1 is not accepted as a fallback direction.

Definition 1.1 (Fixed universal atlas)**.** *For fixed n, the 304 atlas $\mathcal{A}_n$ is the deterministic countable family of cells generated by the following shell families:*

(a) simplex/coordinate cells with one dominant coordinate and balanced tail;

(b) hypercube phase corners with entries in $\{1, i, -1, -i\}$;

(c) golden-projective spiral cells with deterministic amplitudes and phases;

(d) reciprocal or infinity-biased mixed shells with alternating large and small coordinates;

(e) frozen Mobius/Riemann cells whose parameters do not depend on F or the root history.

For a finite trial budget, 304 scans a polynomially bounded window of this atlas. The atlas geometry is input-independent; F only decides whether a tested cell is admissible by predicates.

Definition 1.2 (304 admissible cell predicates)**.** *A tested cell representative y is admissible if:*

(i) $\|F(y)\|$ is finite;

(ii) the logarithmic scale energy of y is below a prescribed bound;

(iii) a fixed local probe set, consisting of radial, phase, and coordinate-simplex probes, contains a point with residual descent by at least a constant factor;

(iv) the normalized equal-value gap between $F(y)$ and the probe values is above a prescribed threshold.

If multiple cells are admissible, 304 uses a cyclic admissible-cell rule. It does not choose the globally smallest residual cell.

Remark 1.3 (Why 304 is cleaner than 301)**.** *The 301 benchmark already showed strong empirical behavior, but its chart parameters react to duplicate pressure, failure pressure, and progress. Those reactions may still be analyzable, but they complicate an invariant distributional proof. 304 removes this complication: the atlas is fixed before F is known. The proof target becomes a geometric measure problem for fixed cells under the Kostlan distribution.*

2 Exact finite-slope algebra

For $\alpha \in \mathbb{N}^n$ write $m_\alpha(z) = z_1^{\alpha_1} \cdots z_n^{\alpha_n}$. For $m \geq 1$ set

$$S_m(u,v) = \sum_{r=0}^{m-1} v^{m-1-r} u^r, \qquad S_0(u,v) = 0.$$

For $a, b \in \mathbb{C}^n$, define

$$q_{\alpha j}(a,b) = \left(\prod_{k<j} b_k^{\alpha_k} \right) S_{\alpha_j}(a_j, b_j) \left(\prod_{k>j} a_k^{\alpha_k} \right).$$

If $F_i(z) = \sum_\alpha c_{i\alpha} m_\alpha(z)$, put

$$Q_F(a,b)_{ij} = \sum_\alpha c_{i\alpha} q_{\alpha j}(a,b).$$

Theorem 2.1 (Exact telescopic finite-slope matrix). *For all $a, b \in \mathbb{C}^n$,*

$$F(b) - F(a) = Q_F(a,b)(b-a).$$

Proof. For a monomial, change coordinates from a to b one coordinate at a time:

$$\prod_{k=1}^n b_k^{\alpha_k} - \prod_{k=1}^n a_k^{\alpha_k} = \sum_{j=1}^n \left(\prod_{k<j} b_k^{\alpha_k} \right) \left(b_j^{\alpha_j} - a_j^{\alpha_j} \right) \left(\prod_{k>j} a_k^{\alpha_k} \right).$$

Since $b_j^m - a_j^m = S_m(a_j, b_j)(b_j - a_j)$, the monomial increment is $\sum_j q_{\alpha j}(a,b)(b_j - a_j)$. Summing over monomials and equations proves the identity. □

Remark 2.2. *$Q_F(a,b)$ is not an approximate Jacobian introduced for convenience. It is an exact polynomial increment factorization. As $b \to a$, $Q_F(a,b) \to J_F(a)$.*

3 Polynomial per-trial cost

Proposition 3.1 (Dense arithmetic cost of one 304 trial). *Assume the trial uses C_A atlas cells, C_P local cell probes, C_E endpoint probes, L line-search points, and S start-balancing candidates/steps, all bounded by* $\mathrm{poly}(n,d,N)$. *Then one 304 trial costs* $\mathrm{poly}(n,d,N)$ *arithmetic operations in the dense representation.*

Proof. A dense evaluation of all n equations over N monomials costs $\mathcal{O}(nN)$ arithmetic operations after monomial powers are organized in the standard dense way, or $\mathcal{O}(ndN)$ under a direct monomial-power implementation; both are polynomial. Testing C_A cells with C_P fixed probes costs $\mathcal{O}(C_A C_P\, ndN)$ up to polynomial bookkeeping. Start balancing with S candidates costs $\mathcal{O}(SndN)$. Building $Q_F(a,b)$ requires, for each equation, variable, and monomial, one telescopic contribution, hence $\mathcal{O}(n^2dN)$ by direct summation, again polynomial. Finite endpoint and curvature probes cost $\mathcal{O}(C_E ndN)$. Solving the $n \times n$ systems for δ_1 and δ_2 costs $\mathcal{O}(n^3)$ by Gaussian elimination. Line search costs $\mathcal{O}(LndN)$. The sum is polynomial whenever the exposed algorithmic budgets are polynomial. □

Remark 3.2 (Benchmark versus proof budgets). *The benchmark used concrete constants: pool* 8192*, epochs* 40*, line-search* 14*, endpoint probe candidates* 12*, universal cells* 16*, universal shells* 5*. These are empirical settings. The complexity theorem only requires that such budgets be bounded by a polynomial if scaled with n, d, N.*

4 Local cubic basin theorem

Let ζ be a regular zero and $e = a - \zeta$. A strict 304 local correction is

$$Q\delta_1 = -F(a), \qquad Q\delta_2 = -\frac{1}{2}H_F(a; b; \delta_1, \delta_1), \qquad a^+ = a + \delta_1 + \delta_2,$$

with $Q = Q_F(a, b)$.

Definition 4.1 (Compatible finite-slope basin)**.** *A ball $B(\zeta, r)$ is a compatible finite-slope basin with constants $(M, L_2, L_3, c_b, c_Q, c_H)$ if, for every accepted endpoint pair (a, b) in the ball:*

(i) $J_F(\zeta)$ is invertible and $\|J_F(z)^{-1}\| \le M$ throughout the ball;

(ii) $\|D^2F(z)\| \le L_2$ and $\|D^3F(z)\| \le L_3$ throughout the ball;

(iii) $\|b - a\| \le c_b \|F(a)\|$;

(iv) $Q_F(a, b)$ is invertible and $\|Q_F(a, b)^{-1}\| \le 2M$;

(v) the finite-slope perturbation is compatible with cubic cancellation:

$$\|(Q_F(a, b) - J_F(a))(a - \zeta)\| \le c_Q \|a - \zeta\|^3 ;$$

(vi) the finite second defect satisfies

$$\left\|H_F(a; b; \delta_1, \delta_1) - D^2F(a)[\delta_1, \delta_1]\right\| \le c_H \|a - \zeta\|^3 .$$

Theorem 4.2 (Local cubic finite-slope convergence)**.** *Inside a compatible finite-slope basin, the strict 304 update satisfies*

$$\left\|a^+ - \zeta\right\| \le C \|a - \zeta\|^3$$

for all a sufficiently close to ζ, where C depends only on the basin constants.

Proof. Taylor expansion at a gives

$$F(a) = J_F(a)e - \frac{1}{2}D^2F(a)[e, e] + \mathcal{O}(\|e\|^3).$$

Let $E_Q = Q_F(a, b) - J_F(a)$. Compatibility gives $E_Qe = \mathcal{O}(\|e\|^3)$, hence $Qe = J_F(a)e + \mathcal{O}(\|e\|^3)$. Since $Q\delta_1 = -F(a)$ and Q^{-1} is bounded,

$$\delta_1 = -e + \frac{1}{2}Q^{-1}D^2F(a)[e, e] + \mathcal{O}(\|e\|^3).$$

Thus $\delta_1 = -e + \mathcal{O}(\|e\|^2)$ and

$$D^2F(a)[\delta_1, \delta_1] = D^2F(a)[e, e] + \mathcal{O}(\|e\|^3).$$

The second-defect assumption gives

$$\delta_2 = -\frac{1}{2}Q^{-1}D^2F(a)[e, e] + \mathcal{O}(\|e\|^3).$$

The quadratic term left after $e+\delta_1$ is therefore cancelled by δ_2, leaving $e+\delta_1+\delta_2 = \mathcal{O}(\|e\|^3)$. $\square$

Remark 4.3 (Compatibility is essential)**.** *The weaker statement $Q_F(a, b) = J_F(a) + \mathcal{O}(\|e\|)$ is insufficient. It can leave an uncancelled quadratic term $J_F(a)^{-1}(Q_F(a, b) - J_F(a))e$. Either the endpoint rule must force the stronger compatibility above, or the finite second defect must include and cancel this transport error.*

5 The exact Smale-17 reduction for 304

Definition 5.1 (304 basin-access probability)**.** *304 has polynomial universal-atlas basin access for Kostlan inputs if there exists a polynomial $P(n,d,N)$ such that one independent 304 trial, using polynomially bounded atlas/probe budgets, reaches a compatible finite-slope basin and validates an approximate zero with probability at least*

$$p_{n,d} \geq \frac{1}{P(n,d,N)}.$$

Theorem 5.2 (Average polynomial time from 304 basin access)**.** *If 304 has polynomial universal-atlas basin access, then repeated independent 304 trials return an approximate zero in expected polynomial arithmetic time over $F \sim \mathcal{K}(n,d)$.*

Proof. By Proposition 3.1, one trial costs at most a polynomial $C(n,d,N)$. The number of independent trials until success is geometric with mean $1/p_{n,d} \leq P(n,d,N)$. The expected cost is therefore at most $C(n,d,N)P(n,d,N)$, polynomial. □

Theorem 5.3 (High-probability amplification)**.** *Under the same basin-access assumption, R independent trials fail with probability at most $\exp(-R/P(n,d,N))$. In particular, $R = P(n,d,N)(\lambda + \log 1/\eta)$ gives failure probability at most $e^{-\lambda}\eta$.*

Proof. The failure probability is $(1-p_{n,d})^R \leq \exp(-p_{n,d}R) \leq \exp(-R/P(n,d,N))$. □

Problem 5.4 (304 universal-atlas basin-access theorem)**.** *Prove that, for $F \sim \mathcal{K}(n,d)$, the fixed 304 universal atlas has inverse-polynomial probability of producing a point in a compatible finite-slope basin of a regular zero, with polynomially bounded predicate and correction budgets. Equivalently, prove that the measure of compatible basins, as seen through the fixed atlas cells and predicates, is not exponentially small in the dense input size.*

Remark 5.5 (Status)**.** *Problem 5.4 is exactly the remaining mathematical obstruction to calling 304 a Smale-17 solution. The benchmark supports the conjecture, and the fixed-atlas design removes an adaptivity complication, but empirical success is not a proof of inverse-polynomial basin measure.*

6 Benchmark 304

The large diagnostic benchmark used the cases

$$(2,2),(2,3),\ldots,(2,15), \quad (3,2),\ldots,(3,8), \quad (4,2),\ldots,(4,5),$$
$$(5,2),(5,3),(5,4),(6,2),(6,3),(7,2),(8,2),(10,2),$$

with seeds $0,\ldots,15$, for 528 runs total. Each run requested $\min(d^n, 50)$ roots. The command-level parameters were

$$\texttt{pool} = 8192, \quad \texttt{epochs} = 40, \quad \texttt{accept} = 10^{-8}, \quad \texttt{tol} = 10^{-12},$$

with line-search 14, endpoint probe candidates 12, universal cells 16, and universal shells 5.

The empirical conclusion is precise: 304 does not recover more requested roots than 301 on this suite, but it produces substantially more alpha-lite certifications and strong roots. Its cost is also substantially higher. Thus 304 is not the speed winner; it is the cleaner and more certifying geometry candidate.

Table 1: Large 304 universal-atlas diagnostic benchmark.

Statistic	Value
Completed runs	528/528
Requested roots recovered	21940/21952
Alpha-lite certified roots	21666/21940
Strong roots	17283/21940
Median seconds/root	$1.0996 \cdot 10^{-1}$
P90 seconds/root	$3.8831 \cdot 10^{-1}$
Median trials/run	379
P90 trials/run	1454
Median residual	$1.0189 \cdot 10^{-13}$
P90 run-median residual	$9.2101 \cdot 10^{-13}$
Median polished residual	$5.2172 \cdot 10^{-15}$
Median cond(J)	4.063

Table 2: 301 versus 304 on the same large suite.

Statistic	301	304
Completed runs	528/528	528/528
Roots recovered	21940/21952	21940/21952
Alpha-lite certified	20549/21940	21666/21940
Strong roots	12054/21940	17283/21940
Median seconds/root	$3.9941 \cdot 10^{-2}$	$1.0996 \cdot 10^{-1}$
P90 seconds/root	$1.3210 \cdot 10^{-1}$	$3.8831 \cdot 10^{-1}$
Median residual	$1.0766 \cdot 10^{-13}$	$1.0189 \cdot 10^{-13}$
Median polished residual	$5.9277 \cdot 10^{-15}$	$5.2172 \cdot 10^{-15}$

7 Conclusion

The 304 engine reduces the non-homotopy Smale-17 route to a single explicit mathematical problem: prove inverse-polynomial basin access for a fixed universal atlas under the Kostlan distribution. Everything after such access is standard and proved here: finite-slope algebra is exact, the local compatible correction is cubic, per-trial cost is polynomial under polynomial budgets, and repetition gives expected polynomial time with high-probability amplification.

The correct claim is therefore conditional but sharp. Benchmark 304 is strong evidence that the fixed atlas sees many useful basins in practice. A Smale-17 theorem still requires a measure-and-conditioning proof for Problem 5.4.